

Vector databases are specialized databases designed to store and search high-dimensional vectors. These vectors are usually embeddings generated by machine learning models.

Vector databases enable semantic search, recommendation systems, and retrieval-augmented generation.

Instead of keyword matching, vector search finds results based on meaning and similarity.

Common vector databases include ChromaDB, FAISS, Pinecone, and Weaviate.

They use approximate nearest neighbor algorithms for fast similarity search.

Vector databases are essential for modern AI applications such as chatbots and document search.

They allow large-scale storage and efficient retrieval of embeddings.